The Secant Method

- A slight variation of Newton's method for functions whose derivatives are difficult to evaluate. $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$
- For these cases the derivative can be approximated by a backward finite divided difference.

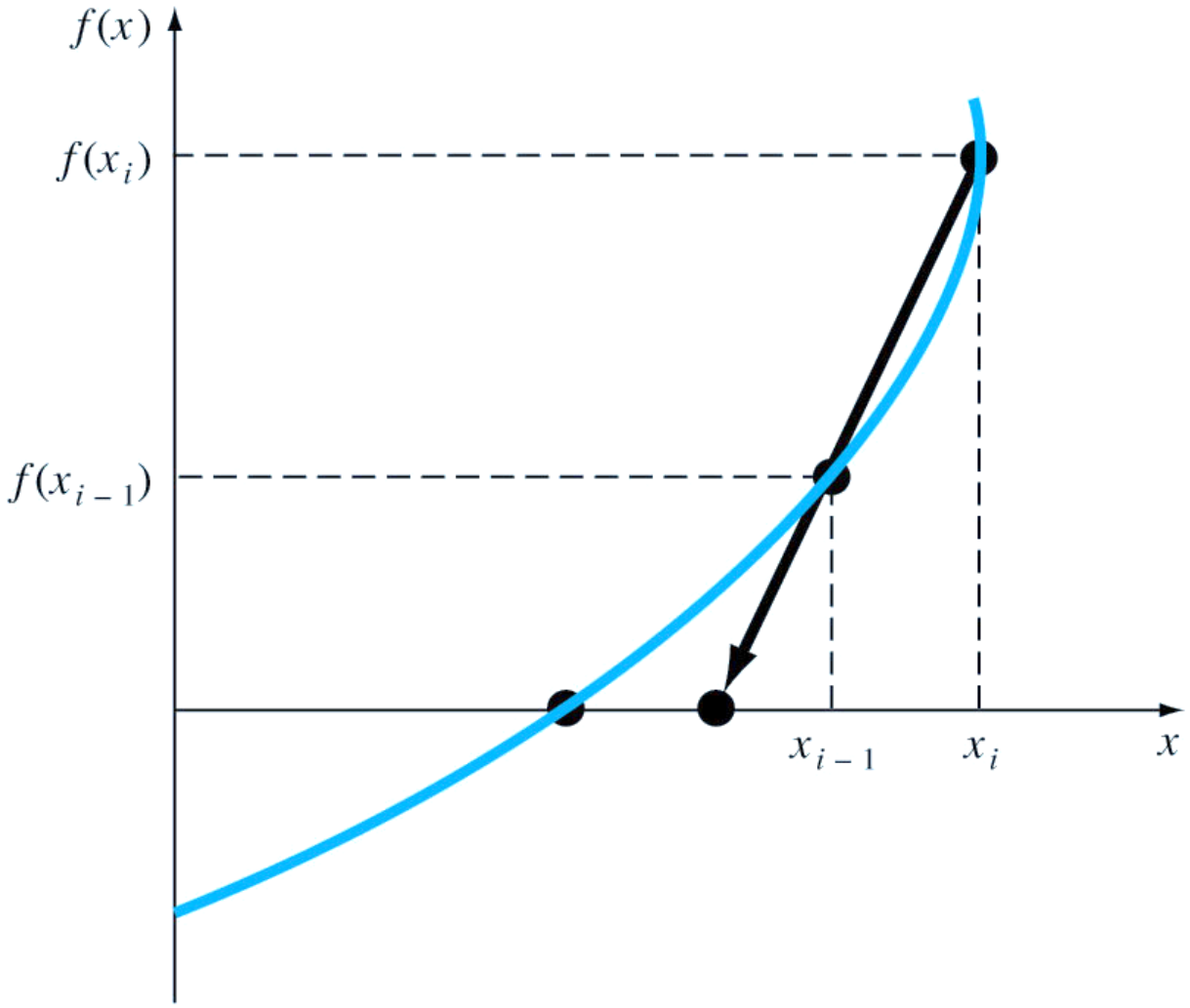
$$\frac{1}{f'(x_i)} \cong \frac{x_i - x_{i-1}}{f(x_i) - f(x_{i-1})}$$

$$x_{i+1} = x_i - f(x_i)\frac{x_i - x_{i-1}}{f(x_i) - f(x_{i-1})} \qquad i = 1,2,3,\ldots$$

- To estimate x_{i+1}, it needs $(x_{i-1}, f(x_{i-1}))$ and $(x_i, f(x_i))$

The Secant Method

- Requires two initial estimates of x, *e.g.*, x_0, x_1. However, because $f(x)$ is not required to change signs between estimates, it is not classified as a "bracketing" method.
- The secant method has the same properties as Newton's method. Convergence is not guaranteed for all x_0 *and* x_1.



Algorithm

Secant method Steps (Rule),

Step-1: Find points x_0 and x_1 such that $x_0<x_1$ and $f(x_0)\cdot f(x_1)<0$.

Step-2: Find next value using

Formula-1 : $x_2 = x_0 - f(x_0)\dfrac{x_1-x_0}{f(x_1)-f(x_0)}$

or Formula-2 : $x_2 = \dfrac{x_0\cdot f(x_1) - x_1\cdot f(x_0)}{f(x_1)-f(x_0)}$

or Formula-3 : $x_2 = x_1 - f(x_1)\dfrac{x_1-x_0}{f(x_1)-f(x_0)}$

(Using any of the formula, we will get same x_2 value)

Algorithm & Example, $f(x) = x^3 - x - 1$

Step-3: If $f(x_2)=0$ *then* x_2 is an exact root,

else $x_0 = x_1$ and $x_1 = x_2$

Step-4: Repeat steps 2 & 3 until $f(x_i)=0$ or

$|f(x_i)| \leq$ Accuracy

Example,

Find a root of an equation $f(x) = x^3 - x - 1$ using Secant method

Solution: List of some x and $f(x)$ values :

x	0	1	2
$f(x)$	-1	-1	5

1st iteration : *Assume* $x_0 = 1$ *and* $x_1 = 2$

Then, $f(x0) = f(1) = -1$ *and* $f(x1) = f(2) = 5$

$$\therefore x_2 = x_0 - f(x_0)\frac{x_1 - x_0}{f(x_1) - f(x_0)}$$

$x_2 = 1 - (-1) \times (2 - 1)/(5 - (-1))$

$x_2 = 1.16667$

$\therefore f(x_2) = f(1.16667) = -0.5787$

Example,

<u>2nd *iteration*</u> : $x_1=2$ and $x_2=1.16667$

$f(x_1) = f(2) = 5$ and $f(x_2) = f(1.16667) = -0.5787$

$$\therefore x_3 = x_1 - f(x_1)\frac{x_2 - x_1}{f(x_2) - f(x_1)}$$

$x_3=2 - 5\times(1.16667-2)/(-0.5787-5)$

$x_3=1.25311$

$\therefore f(x_3)=f(1.25311)=-0.28536$

Example,

3rd iteration : $x_2=1.16667$ and $x_3=1.25311$

$f(x_2)=f(1.16667)=-0.5787$

and $f(x_3)=f(1.25311)=-0.28536$

$$\therefore x_4 = x_2 - f(x_2)\frac{x_3-x_2}{f(x_3)-f(x_2)}$$

$x_4=1.16667-(-0.5787)\times(1.25311-1.16667)/$

$(-0.28536-(-0.5787))$

$x_4=1.33721$

$\therefore f(x_4)=f(1.33721)=0.05388$

Example,

5th iteration : $x_4=1.33721$ and $x_5=1.32385$

$f(x_4)=f(1.33721)=0.05388$ and

$f(x_5)=f(1.32385)=-0.0037$

$$\therefore x_6 = x_4 - f(x_4)\frac{x_5-x_4}{f(x_5)-f(x_4)}$$

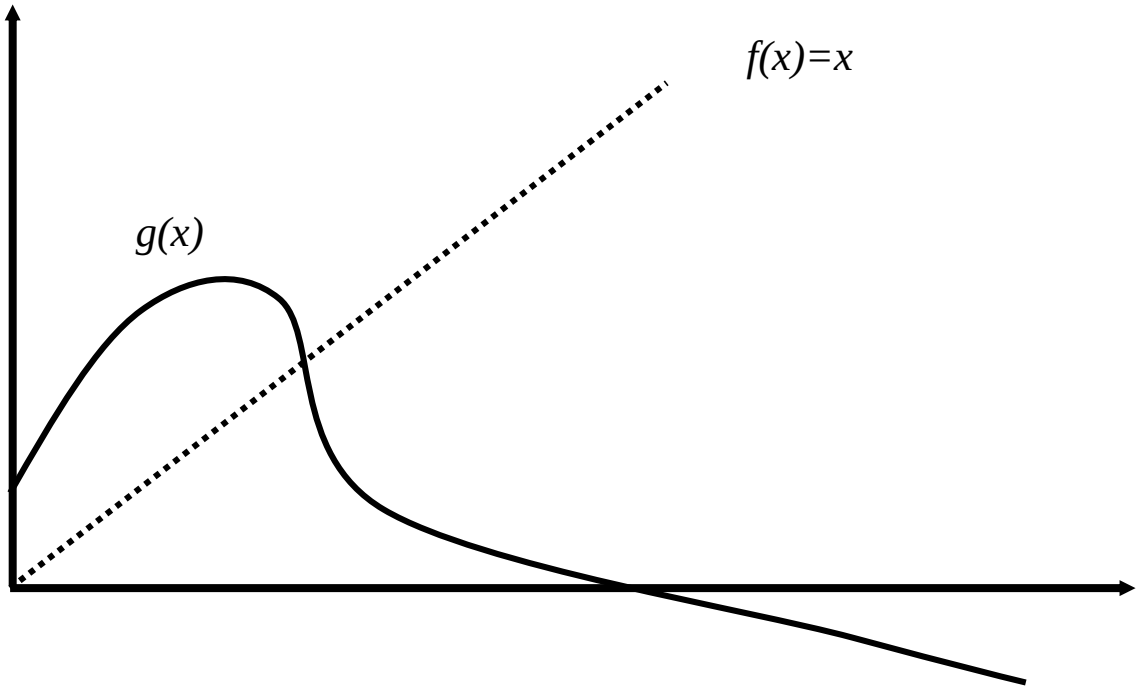
$x_6=1.33721-0.05388\times(1.32385-1.33721)/(-0.0037-0.05388)$

$x_6=1.32471$

$\therefore f(x_6)=f(1.32471)=-0.00004$

Fixed-Point Iteration---- Successive Approximation

- Many problems also take on the specialized form: $g(\boldsymbol{x})=\boldsymbol{x}$, where we seek, $\boldsymbol{x}$, that satisfies this equation.
- In the limit, $f(x_k)=0$, hence $x_{k+1}=x_k$



Simple Fixed-Point Iteration

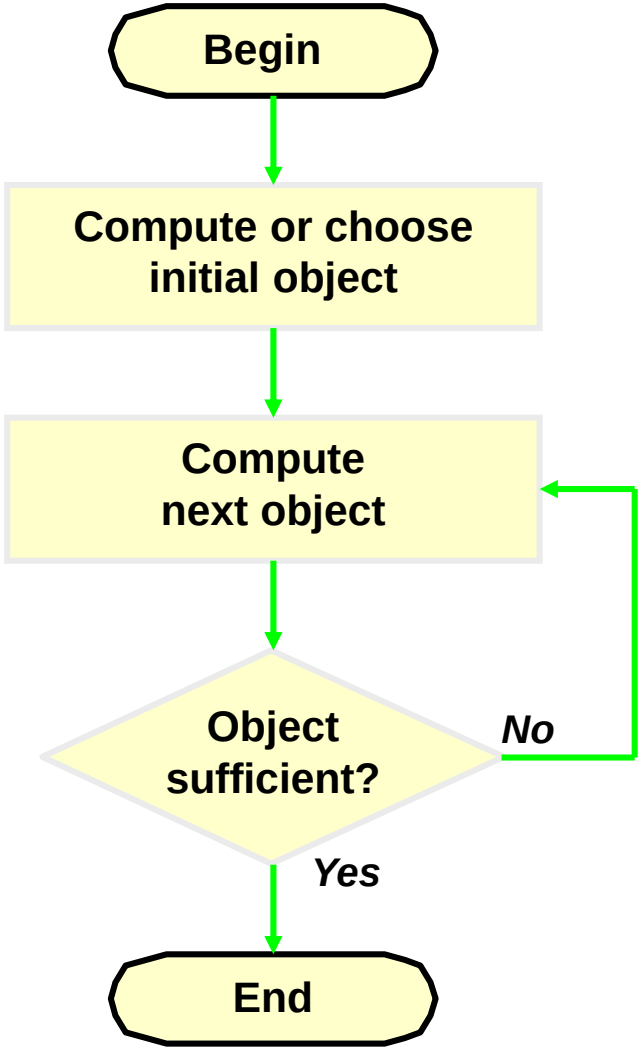
- Rearrange the function $f(x) = 0$ so that x is on the left-hand side of the equation: $x = g(x)$
- Use the new function g to predict a new value of x - that is,

$$x_{i+1} = g(x_i)$$

- The approximate error is given by:

$$\varepsilon_a = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| 100\%$$

Successive Approximations



Example:

$$f(x) = x^2 - x - 2 \qquad x > 0$$

$$g(x) = x^2 - 2$$

$$or$$

$$g(x) = \sqrt{x + 2}$$

$$or$$

$$g(x) = 1 + \frac{2}{x}$$

Iterative Solution

Find the root of

$f(x) = e^{-x} - x$ $\qquad x_{n+1} = e^{-x_n}$

1. Start with a guess say $x_1=1$,
2. Generate

   a) $x_2 = e^{-x1} = e^{-1} = 0.368$

   b) $x_3 = e^{-x2} = e^{-0.368} = 0.692$

   c) $x_4 = e^{-x3} = e^{-0.692} = 0.500$

n	x_n
1	1.000
2	0.368
3	0.692
4	0.500
5	0.606
6	0.545
7	0.579
8	0.560
9	0.571
10	0.564
11	0.568
12	0.566
13	0.567
14	0.567
15	0.567

In general:

After a few more iteration we will get $0.567 \approx e^{-0.567}$

Problem $f(x) = 2x^2 - 4x + 1$

- Find a root near *x=1.0* and *x=2.0*
- Solution: $x = g(x) = \frac{1}{2}x^2 + \frac{1}{4}$
  - Starting at *x=1*, *x=0.292893* at *15th* iteration
  - Starting at *x=2*, it will not converge
  - Why? Relate to *g'(x)=x*. for convergence *g'(x) < 1*

$$x = g(x) = \sqrt{2x - \tfrac{1}{2}}$$

  - Starting at x=1, x=1.707 at iteration 19
  - Starting at x=2, x=1.707 at iteration 12
  - Why? Relate to $g'(x) = \left(2x - \tfrac{1}{2}\right)^{-\frac{1}{2}}$

Examples

If $f(x) = x^2 - x - 2$, then fixed points of each of functions

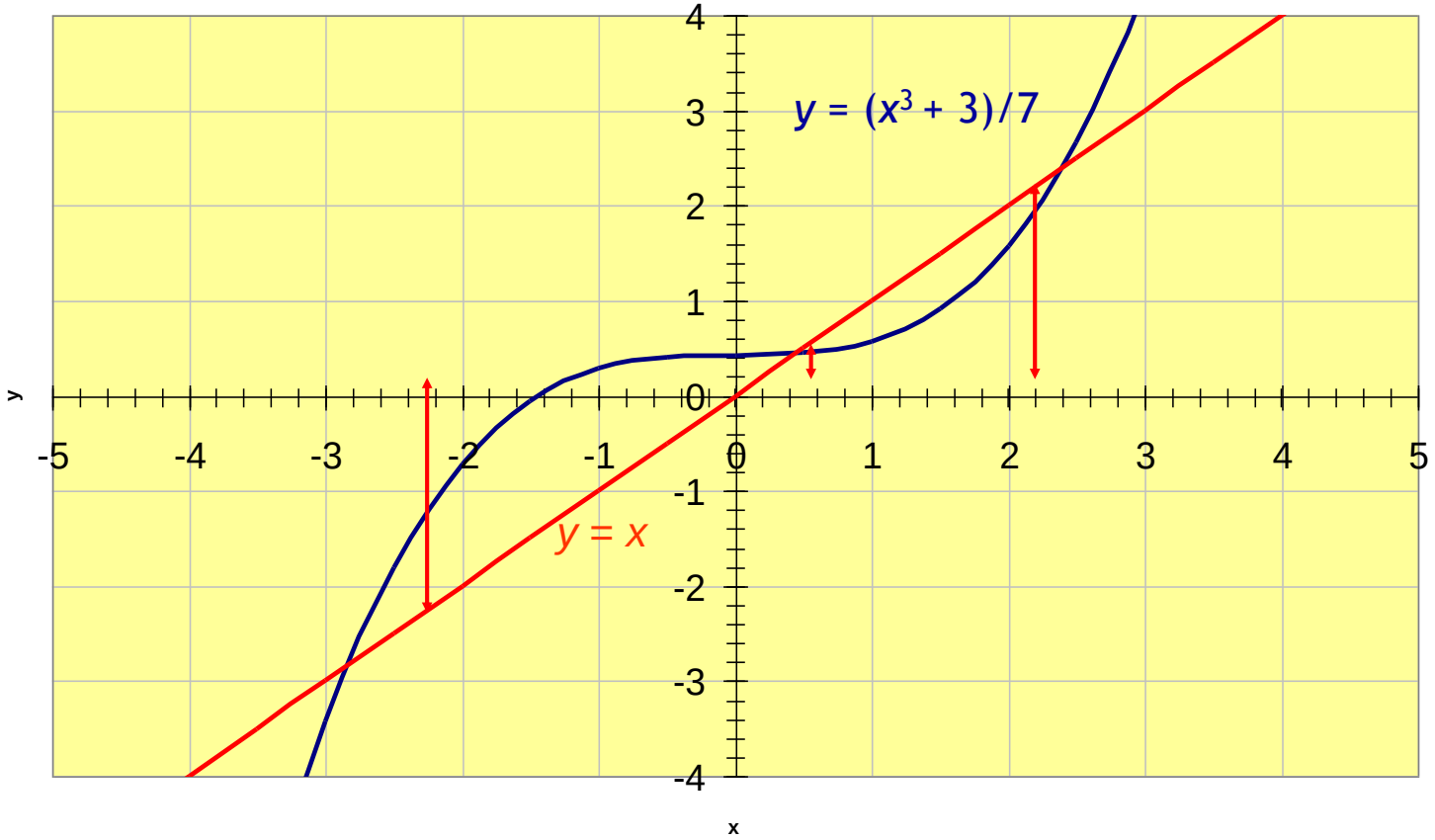
- $g(x) = x^2 - 2$
- $g(x) = \sqrt{x+2}$
- $g(x) = 1 + 2/x$
- $g(x) = \dfrac{x^2+2}{2x-1}$

are solutions to equation $f(x) = 0$

Fixed Point Iteration

The equation $f(x) = 0$, where $f(x) = x^3 - 7x + 3$, may be re-arranged to give $x = (x^3 + 3)/7$.

Intersection of the graphs of $y = x$ and $y = (x^3 + 3)/7$ represent roots of the original equation $x^3 - 7x + 3 = 0$.



Fixed Point Iteration

The rearrangement $x = (x^3 + 3)/7$ leads to the iteration

$$x_{n+1} = \frac{x_n^{\ 3} + 3}{7}, \quad n = 0, 1, 2, 3, \ldots$$

To find the middle root a, let initial approximation $x_0 = 2$.

$$x_1 = \frac{x_0^{\ 3} + 3}{7} = \frac{2^3 + 3}{7} = 1.57143$$

$$x_2 = \frac{x_1^{\ 3} + 3}{7} = \frac{1.57143^3 + 3}{7} = 0.98292$$

$$x_3 = \frac{x_2^{\ 3} + 3}{7} = \frac{0.98292^3 + 3}{7} = 0.56423$$

$$x_4 = \frac{x_3^{\ 3} + 3}{7} = \frac{0.56423^3 + 3}{7} = 0.45423 \qquad \text{etc.}$$

The iteration slowly converges to give a = **0.441** (to 3 s.f.)

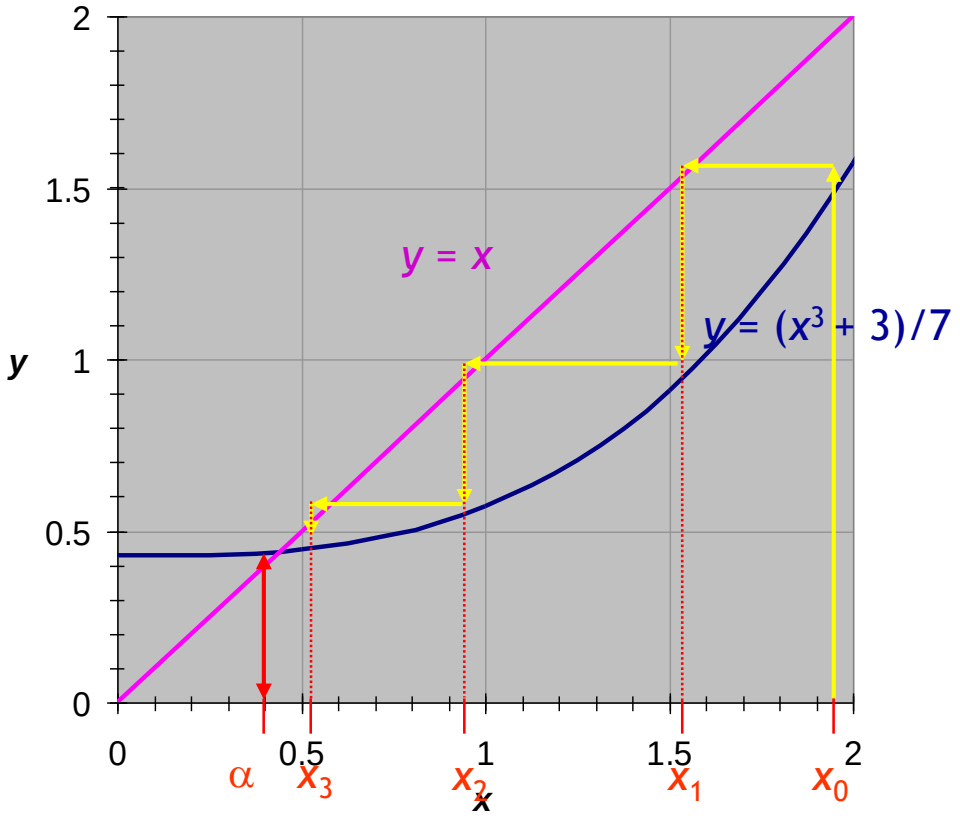
Fixed Point Iteration

The rearrangement $x = (x^3 + 3)/7$ leads to the iteration

$$x_{n+1} = \frac{x_n^{\,3} + 3}{7}, \quad n = 0, 1, 2, 3, \ldots$$

For $x_0 = 2$ the iteration will converge on the middle root a, since $g'(a) < 1$.

n	x_n
0	2
1	1.57143
2	0.98292
3	0.56423
4	0.45423
5	0.44196
6	0.4409
7	0.44082
8	0.44081



α = **0.441** (to 3 s.f.)

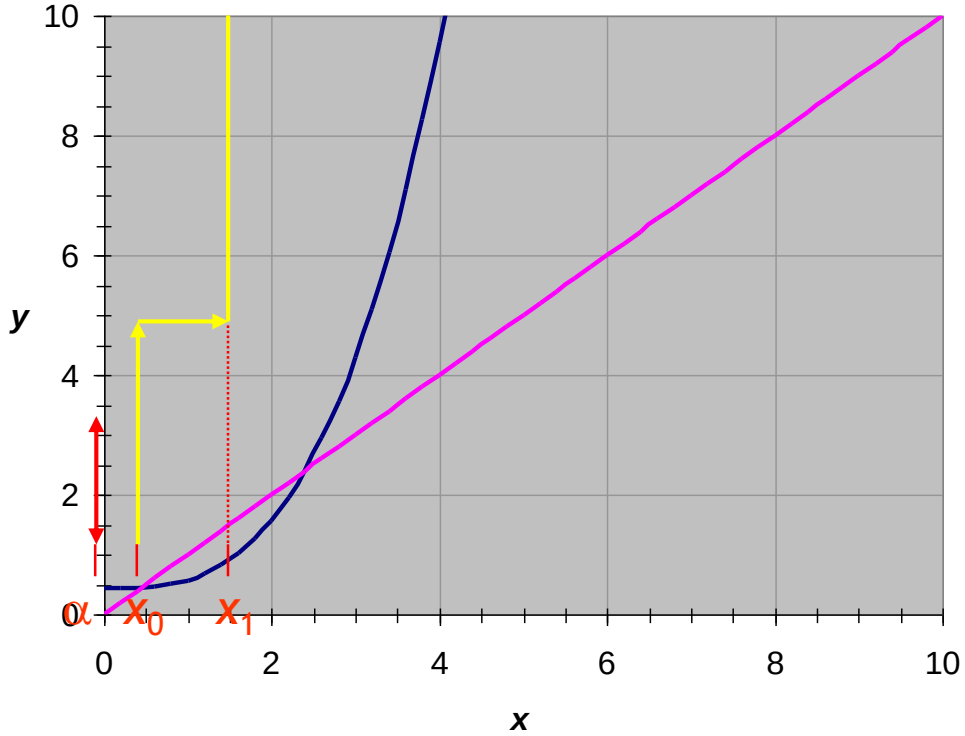
Fixed Point Iteration - breakdown

The rearrangement $x = (x^3 + 3)/7$ leads to the iteration

$$x_{n+1} = \frac{x_n^{\,3} + 3}{7}, \quad n = 0, 1, 2, 3, \ldots$$

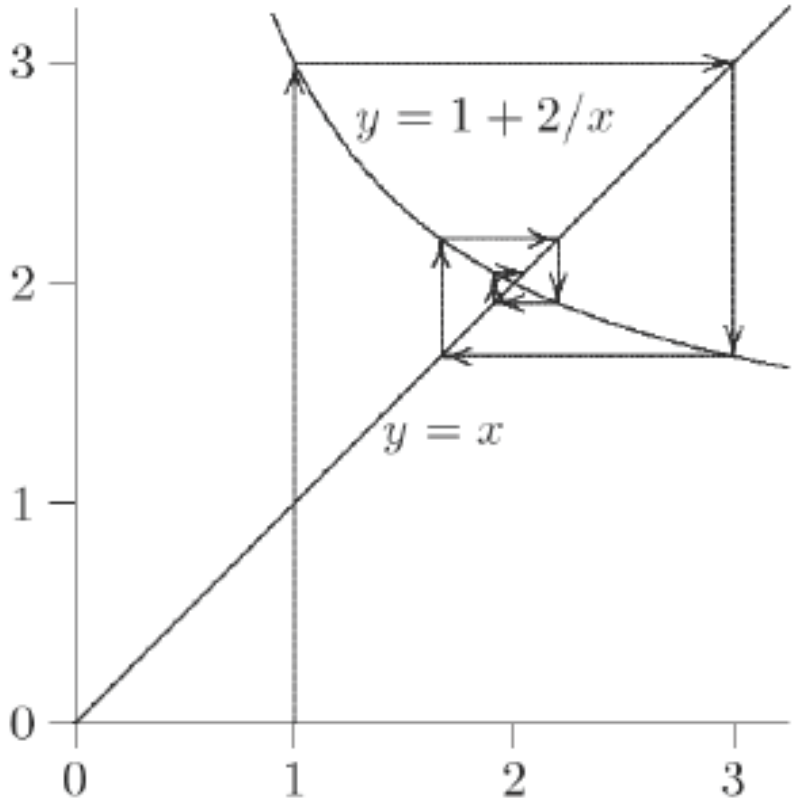
For $x_0 = 3$ the iteration will diverge from the upper root a.

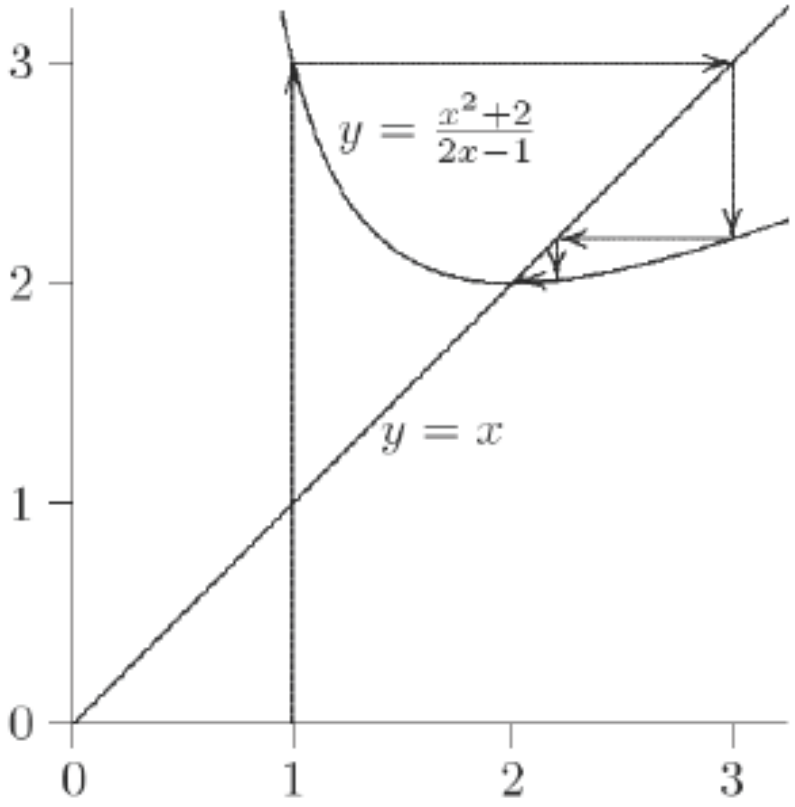
n	x_n
0	3
1	4.28571
2	11.6739
3	227.702
4	1686559
5	6.9E+17



The iteration diverges because $g'(a) > 1$.

Example: FPI





Convergence of FPI

- If $x^* = g(x^*)$ and $|g'(x^*)| < 1$, then there is interval containing x^* such that iteration

$$x_{k+1} = g(x_k)$$

  converges to x^* if started within that interval

- If $|g'(x^*)| > 1$, then iterative scheme diverges
- Asymptotic convergence rate of fixed-point iteration is usually linear, with constant $C = |g'(x^*)|$
- But if $g'(x^*) = 0$, then convergence rate is at least quadratic

Simple Fixed-Point Iteration Convergence

$x_r - x_{i+1} = g(x_r) - g(x_i)$ [*slope =1*]

Let $a = x_i$ and $b = x_r$

$$g'(\xi) = \frac{g(x_r) - g(x_i)}{x_r - x_i}$$

$$g(x_r) - g(x_i) = (x_r - x_i)g'(\xi)$$

$$then\ x_r - x_{i+1} = (x_r - x_i)g'(\xi)$$

$$\varepsilon_{a,i+1} = g'(\xi)\varepsilon_{a,i}$$

$If |g'(\xi)| < 1.0$ the error decreases with each iteration

$If |g'(\xi)| > 1.0$ the error increases with each iteration

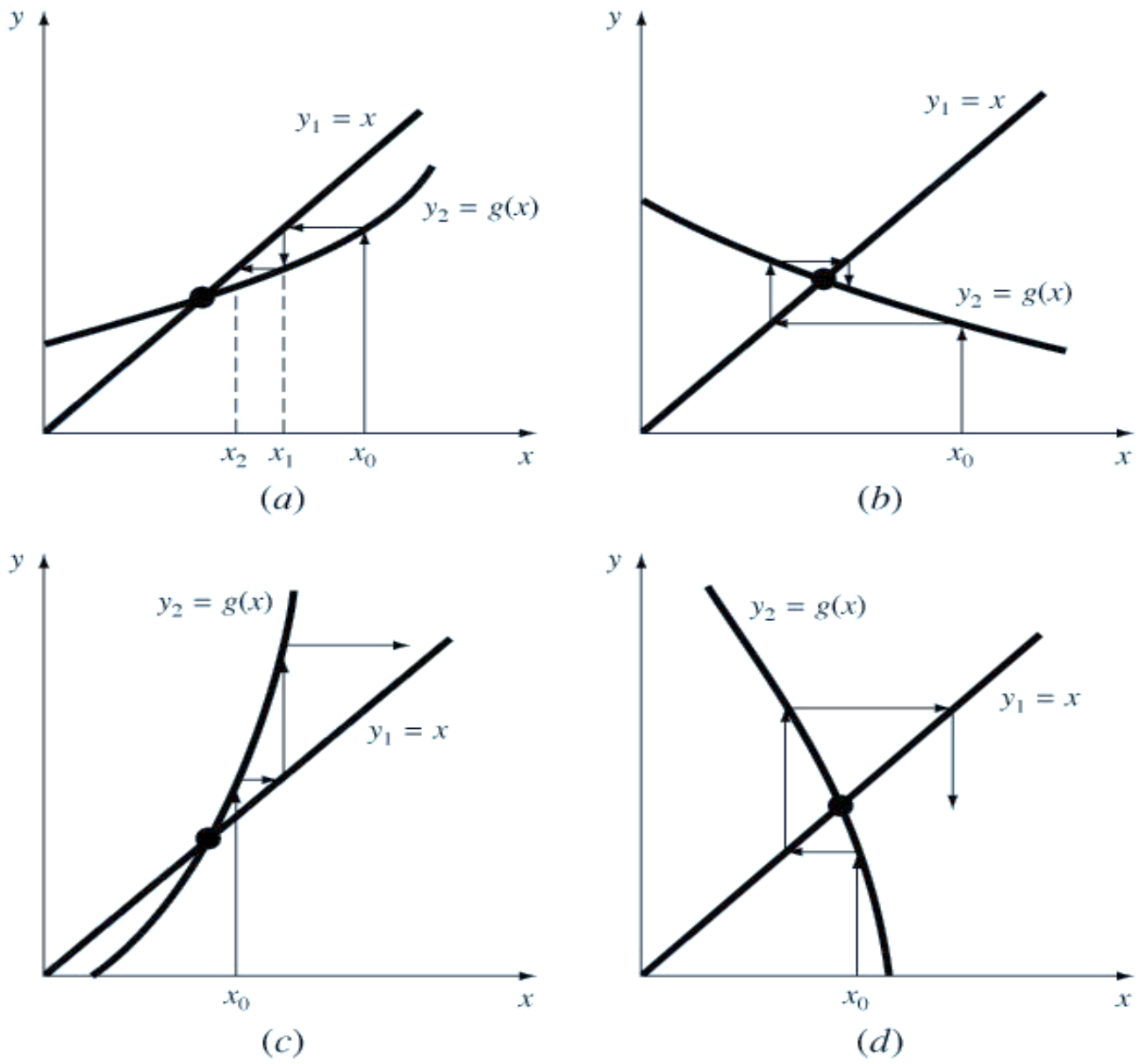
Simple Fixed-Point Iteration Convergence

- Fixed-point iteration converges if :

$$|g'(x)| < 1 \quad \text{(slope of the line } f(x) = x\text{)}$$

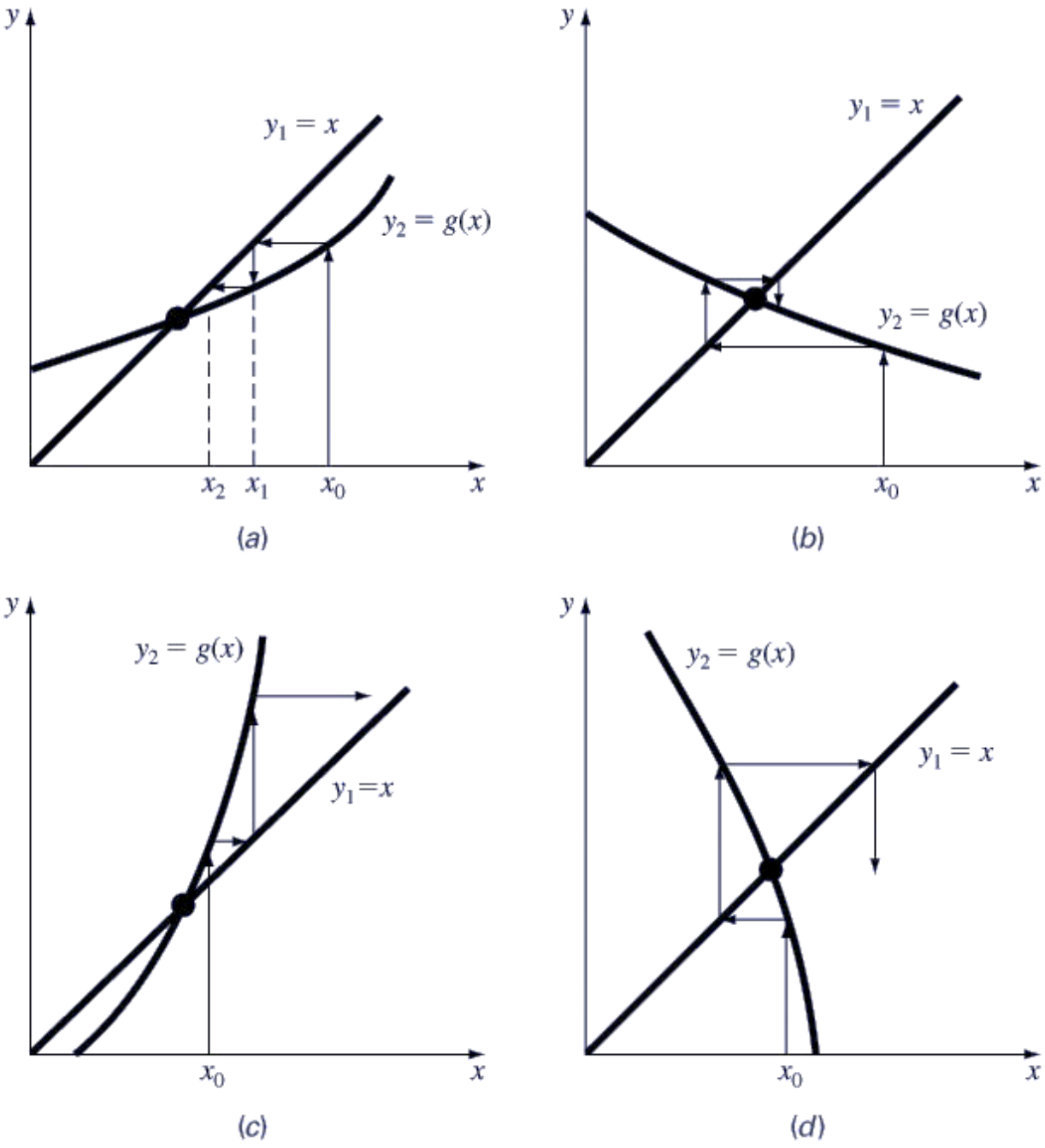
- When the method converges, the error is roughly proportional to or less than the error of the previous step, therefore it is called “linearly convergent.”

Simple Fixed-Point Iteration-Convergence



More on Convergence

- Graphically→ the solution is at the intersection of the two curves. We identify the point on y_2 corresponding to the initial guess and the next guess corresponds to the value of the argument x where $y_1(x) = y_2(x)$.
- Convergence of the simple fixed-point iteration method requires that the derivative of $g(x)$ near the root has a magnitude less than 1.
  a) Convergent, $0 \leq g' < 1$
  b) Convergent, $-1 < g' \leq 0$
  c) Divergent, $g' > 1$
  d) Divergent, $g' < -1$



Fixed Point Iteration

1. Use an initial guess $x = 1.5$ and $y = 3.5$
2. The iteration formulae:

   $x_{i+1} = (10 - x_i^2)/y_i$ and $y_{i+1} = 57 - 3x_i y_i^2$
3. <u>First iteration,</u>

   $x = (10-(1.5)^2)/3.5 = 2.21429$

   $y = (57-3(2.21429)(3.5)^2 = -24.37516$
4. <u>Second iteration:</u>

   $x = (10-2.21429^2)/-24.37516 = -0.209$

   $y = 57-3(-0.209)(-24.37516)^2 = 429.709$
5. Solution is **diverging** so try another iteration formula

Examples

Determine the lowest positive root of:

$$f(x) = 8\, e^{-x} \sin(x) - 1$$

Using the **Newton-Raphson method** (three iterations, $x_0 = 0.3$)

Summary

Method	Pros	Cons
Bisection	- Easy, Reliable, Convergent - One function evaluation per iteration - No knowledge of derivative is needed	- Slow - Needs an interval [a,b] containing the root, i.e., $f(a)f(b)<0$
Newton	- Fast (if near the root) - Two function evaluations per iteration	- May diverge - Needs derivative and an initial guess x_0 such that $f'(x_0)$ is nonzero
Secant	- Fast (slower than Newton) - One function evaluation per iteration - No knowledge of derivative is needed	- May diverge - Needs two initial points guess x_0, x_1 such that $f(x_0)- f(x_1)$ is nonzero